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Inventor(s)	Zeng; Yizhi

Method of manufacturing semiconductor structure and semiconductor structure

Abstract

The present disclosure provides a method of manufacturing a semiconductor structure and a semiconductor structure, and relates to the technical field of semiconductors. The method of manufacturing a semiconductor structure includes: providing a base; forming a first dielectric layer having grooves, where the grooves include a first groove and a second groove whose inner diameters have a first difference; forming an initial filling layer in the first groove and the second groove, where the first groove forms a third groove, the second groove forms a fourth groove, and an inner diameter of the third groove and that of the fourth groove have a second difference less than the first difference; processing the initial filling layer to form a filling layer; and forming a plurality of etching openings whose inner diameters have a preset difference on a surface of the base.

Inventors:	Zeng; Yizhi (Hefei, CN)
Applicant:	CHANGXIN MEMORY TECHNOLOGIES, INC. (Hefei, CN)
Family ID:	83036338
Assignee:	CHANGXIN MEMORY TECHNOLOGIES, INC. (Hefei, CN)
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Primary Examiner: Duclair; Stephanie P

Attorney, Agent or Firm: Kilpatrick Townsend & Stockton LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority of Chinese Patent Application No. 202210703726.6, submitted to the Chinese Intellectual Property Office on Jun. 21, 2022, the disclosure of which is incorporated herein in its entirety by reference.

TECHNICAL FIELD

(2) The present disclosure relates to the technical field of semiconductors, and in particular, to a method of manufacturing a semiconductor structure and a semiconductor structure.

BACKGROUND

(3) A dynamic random access memory (DRAM) is a semiconductor memory that randomly writes and reads data at a high speed, and is widely used in data storage devices or apparatuses.

(4) With the continuous development of DRAM process technologies, a feature dimension, such as a critical dimension (CD) of an active region, of a semiconductor structure in an integrated circuit, is also decreasing. In addition, when semiconductor structures with smaller CDs are manufactured, local CD uniformity (LCDU) is excessively low, adversely affecting the electrical performance of

the semiconductor structures.

SUMMARY

(5) An overview of the subject matter detailed in the present disclosure is provided below, which is not intended to limit the protection scope of the claims.

(6) The present disclosure provides a method of manufacturing a semiconductor structure and a semiconductor structure.

(7) A first aspect of the present disclosure provides a method of manufacturing a semiconductor structure. The method of manufacturing a semiconductor structure includes: providing a base; forming a first dielectric layer on the base, where the first dielectric layer is provided with grooves arranged in an array, the grooves include a first groove and a second groove, and an inner diameter of the first groove and that of the second groove have a first difference; forming an initial filling layer, where the initial filling layer covers a sidewall and a bottom wall of the first groove and covers a sidewall and a bottom wall of the second groove, the initial filling layer in the first groove forms a third groove, the initial filling layer in the second groove forms a fourth groove, an inner diameter of the third groove and that of the fourth groove have a second difference, and the second difference is less than the first difference; processing the initial filling layer to form a filling layer; and etching the filling layer and/or the first dielectric layer, to form a plurality of etching openings whose inner diameters have a preset difference on a surface of the base.

(8) A second aspect of the present disclosure provides a semiconductor structure, which is manufactured by using the method of manufacturing a semiconductor structure according to any one of the foregoing embodiments.

(9) Other aspects of the present disclosure are understandable upon reading and understanding of the accompanying drawings and detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings incorporated into the specification and constituting a part of the specification illustrate the embodiments of the present disclosure, and are used together with the description to explain the principles of the embodiments of the present disclosure. In these accompanying drawings, similar reference numerals represent similar elements. The accompanying drawings in the following description illustrate some rather than all of the embodiments of the present disclosure. Those skilled in the art may obtain other accompanying drawings based on these accompanying drawings without creative efforts.

(2) FIG. 1 is a flowchart of a method of manufacturing a semiconductor structure according to an exemplary embodiment;

(3) FIG. 2 is a schematic diagram of forming a first dielectric layer in a method of manufacturing a semiconductor structure according to an exemplary embodiment;

(4) FIG. 3 is a top view of a first mask pattern in a method of manufacturing a semiconductor structure according to an exemplary embodiment;

(5) FIG. 4 is a schematic diagram of forming a fifth groove and a sixth groove in a method of manufacturing a semiconductor structure according to an exemplary embodiment;

(6) FIG. 5 is a schematic diagram of forming a first groove and a second groove in a method of manufacturing a semiconductor structure according to an exemplary embodiment;

(7) FIG. 6 is a schematic diagram of forming an initial filling layer in a method of manufacturing a semiconductor structure according to an exemplary embodiment;

(8) FIG. 7 is a schematic diagram of performing ion implantation on an initial filling layer in a method of manufacturing a semiconductor structure according to an exemplary embodiment;

(9) FIG. 8 is a schematic diagram of performing RTP on an initial filling layer in a method of

- manufacturing a semiconductor structure according to an exemplary embodiment;
- (10) FIG. 9 is a schematic diagram of forming an etching opening in a method of manufacturing a semiconductor structure according to an exemplary embodiment;
- (11) FIG. 10 is a top view of FIG. 9;
- (12) FIG. 11 is a schematic diagram of forming a first through hole in a method of manufacturing a semiconductor structure according to an exemplary embodiment;
- (13) FIG. 12 is a schematic diagram of forming a second through hole in a method of manufacturing a semiconductor structure according to an exemplary embodiment; and
- (14) FIG. 13 is a schematic diagram of forming a contact hole in a method of manufacturing a semiconductor structure according to an exemplary embodiment.

DETAILED DESCRIPTION

(15) To make the objectives, technical solutions, and advantages of the embodiments of the present disclosure clearer, the following clearly and completely describes the technical solutions in the embodiments of the present disclosure with reference to the accompanying drawings in the embodiments of the present disclosure. Apparently, the described embodiments are some rather than all of the embodiments of the present disclosure. All other embodiments obtained by those skilled in the art based on the embodiments of the present disclosure without creative efforts should fall within the protection scope of the present disclosure. It should be noted that the embodiments in the present disclosure and features in the embodiments may be combined with each other in a non-conflicting manner.

(16) A DRAM is a semiconductor memory that randomly writes and reads data at a high speed, and is widely used in data storage devices or apparatuses.

(17) With the continuous development of DRAM process technologies, a feature dimension, such as a CD of an active region, of a semiconductor structure in an integrated circuit, is also shrinking. In addition, when semiconductor structures with smaller CDs are manufactured, LCDU is excessively low, leading to different local CDs of subsequently formed semiconductor structures. In a process of forming through holes (where the through holes may form a contact structure, a capacitor structure, or the like) in the semiconductor structure, sizes of mask patterns formed in a mask are not exactly the same, and the uniformity is poor. The sizes are normally distributed by the limitation of a mask manufacturing process. Sizes of most of mask holes in the mask pattern satisfy a requirement of a subsequent etch processing. However, some of the mask holes have width inner diameters or inner diameter sizes that greatly differ from a target size, and such mask holes may lead to the formation of laps between the subsequently formed semiconductor structures (such as bit line contacts), adversely affecting the electrical performance of the semiconductor structures.

(18) To resolve one of the foregoing technical problems, an exemplary embodiment of the present disclosure provides a method of manufacturing a semiconductor structure. The following describes the method of manufacturing a semiconductor structure with reference to FIG. 1 to FIG. 13.

(19) The semiconductor structure is not limited in this embodiment. The semiconductor structure is described below by using a DRAM as an example, but this embodiment is not limited thereto, and the semiconductor structure in this embodiment may also be other structures.

(20) As shown in FIG. 1, an exemplary embodiment of the present disclosure provides a method of manufacturing a semiconductor structure, including:

(21) Step S100: Provide a base.

(22) Step S200: Form a first dielectric layer on the base, where the first dielectric layer is provided with grooves arranged in an array, the grooves include a first groove and a second groove, and an inner diameter of the first groove and that of the second groove have a first difference.

(23) Step S300: Form an initial filling layer, where the initial filling layer covers a bottom wall and a sidewall of the first groove and covers a sidewall and a bottom wall of the second groove, the first groove forms a third groove, the second groove forms a fourth groove, an inner diameter of the third groove and that of the fourth groove have a second difference, and the second difference is

less than the first difference.

(24) Step **S400**: Process the initial filling layer to form a filling layer.

(25) Step **S500**: Etch the filling layer and/or the first dielectric layer, to form a plurality of etching openings whose inner diameters have a preset difference on a surface of the base.

(26) As shown in FIG. 2, in step **S100**, a base **10** serves as a support component of a DRAM and is configured to support other components provided thereon. The base **10** may be made of a semiconductor material. The semiconductor material may be one or more of silicon, germanium, a silicon-germanium compound, and a silicon-carbon compound. In this embodiment, the base **10** is made of silicon, to facilitate understanding of a subsequent forming method by those skilled in the art, rather than to constitute a limitation. In an actual application process, an appropriate material of the base may be selected as required.

(27) Referring to FIG. 2, in step **S200**, a first dielectric layer **20** may be formed on a base **10** by using an atomic layer deposition process, a physical vapor deposition process, or a chemical vapor deposition process. A material of the first dielectric layer **20** may include but is not limited to silicon oxynitride or undoped polycrystalline silicon. To form another semiconductor structure (such as a bit line structure or a word line structure) on the first dielectric layer **20**, an opening (not shown in the figure) needs to be formed on the first dielectric layer **20**. For example, a mask layer (not shown in the figure), a photoresist (not shown in the figure), and the like are formed through deposition on the first dielectric layer **20**, and then a plurality of openings arranged in an array are formed on the first dielectric layer **20** by using an etching processing process, where the first dielectric layer **20** between adjacent openings forms a mask structure **30**.

(28) Referring to FIG. 5, with reference to FIG. 4, grooves **31** arranged in an array are provided on the first dielectric layer **20**. It should be noted that, a region between two adjacent mask structures **30** along a same row and/or a same column is a groove **31**. The grooves **31** include a first groove **40** and a second groove **50**. The first groove **40** may be a region between two adjacent mask structures **30** of some mask structures **30** in a same row and/or a same column, and the second groove **50** may be a region between two adjacent mask structures **30** of some mask structures **30** in a same row and/or a same column, where the first groove **40** and the second groove **50** are distributed in different regions on the first dielectric layer **20**.

(29) A plurality of mask structures **30** in a same column are defined to be provided at intervals along a first direction X. Referring to FIG. 2, an orientation shown in the figure is used as an example. The first direction is the direction X in FIG. 2, that is, the first direction X is parallel to a front side surface of the base **10**. There may be a plurality of first grooves **40** and a plurality of second grooves **50** along the first direction X. In a process of forming the mask structures **30**, sizes of the mask structures **30** along the first direction X are different, and therefore, an inner diameter of the first groove **40** is different from an inner diameter of the second groove **50**. The inner diameter of the first groove **40** may be greater than or less than the inner diameter of the second groove **50**.

(30) The inner diameter may be a size, of the groove **31**, measured in a same direction and is not limited to the first direction X and the second direction in FIG. 3, and for all of the grooves **31**, diameters obtained through measurement in a same direction are inner diameters of the grooves **31**. It should be noted that, the second direction is a direction Y in FIG. 3, that is, the second direction Y and the first direction X are perpendicular to each other, and the second direction Y is parallel to a left side surface or a right side surface of the base **10**.

(31) The inner diameter of the first groove **40** satisfies a requirement on a subsequently etching processing process; or the inner diameter of the second groove **50** satisfies a requirement on a subsequently etching processing process. In this embodiment and the embodiments described below, that the inner diameter of the second groove **50** is greater than the inner diameter of the first groove **40** is used as an example for description, and the inner diameter of the second groove **50** and the inner diameter of the first groove **40** have a first difference.

(32) In another embodiment, referring to FIG. 3, the first dielectric layer **20** may further include a third initial groove, a fourth initial groove, and the like whose inner diameters are different from the inner diameter of the first groove **40** and that of the second groove **50** along the first direction X.

(33) Referring to FIG. 6, in step S300, the initial filling layer **61** may be formed through deposition in the first groove **40** and the second groove **50** by using an atomic layer deposition process, a chemical vapor deposition process, or a physical vapor deposition process. The initial filling layer **61** is deposited on a bottom wall and a sidewall of the first groove **40** and those of the second groove **50**, and the initial filling layer **61** further covers a top surface of the mask structure **30**.

(34) It should be noted that, in the process of forming the initial filling layer **61** by using the deposition processes, a deposition parameter (such as a deposition rate, deposition time, or particle deposition energy) of a deposition material used for forming the initial filling layer **61** may be controlled, such that the deposition material used for forming the initial filling layer **61** forms part of the initial filling layer **61** whose deposition thickness is relatively thick on the bottom wall and the sidewall of the second groove **50**, and forms part of the initial filling layer **61** whose deposition thickness is relatively thin on the bottom wall and the sidewall of the first groove **40**.

(35) After the initial filling layer **61** is formed through deposition, the initial filling layer **61** in the first groove **40** forms a third groove **70**, and the initial filling layer **61** in the second groove **50** forms a fourth groove **80**.

(36) The inner diameter of the third groove **70** and the inner diameter of the fourth groove **80** have a second difference along the first direction X, where the second difference is less than the first difference. After the third groove **70** and the fourth groove **80** are formed, the inner diameter of the fourth groove **80** may be greater than the inner diameter of the third groove **70** or may be less than or equal to that of the third groove **70** along the first direction X.

(37) Referring to FIG. 7 and FIG. 8, in step S400, the initial filling layer **61** is processed, to form a filling layer **60**, so as to increase a rate for subsequent etching and reduce a production period of a semiconductor structure.

(38) Referring to FIG. 9 and FIG. 10, in step S500, the filling layer **60** and/or the first dielectric layer **20** are etched by using an etching processing process, to form a plurality of etching openings **90** whose inner diameters have a preset difference on a surface of the base **10**. It should be noted that, the preset difference may be a difference satisfying a requirement on a subsequent etching processing process or an error operation range, and the preset difference may be 1 nm.

(39) In this embodiment, a spacing between adjacent grooves is patched by using the initial filling layer, and the initial filling layer is processed, such that any subsequently formed etching opening can satisfy an etching process requirement, thereby effectively improving LCDU and an electrical property of the semiconductor structure.

(40) Referring to FIG. 3, in some embodiments, the groove **31** has a target size, and the target size may be understood as a size satisfying the etching process requirement. The inner diameter of the first groove **40** is less than the target size, and the inner diameter of the second groove **50** is greater than the target size.

(41) The target size is any value in a range of 30 nm to 60 nm. For example, the target size may be 30 nm, 35 nm, 40 nm, 45 nm, 50 nm, 55 nm, 60 nm, or the like. The inner diameter of the first groove **40** and that of the second groove **50** have a maximum difference of 6 nm, which is a difference allowable in a mask manufacturing process. If the maximum difference is greater than the difference, a manufactured mask does not satisfy a subsequent process requirement because the inner diameters have an excessively large difference.

(42) Referring to FIG. 7 and FIG. 8, in some embodiments, the initial filling layer **61** may be processed by using the following method.

(43) As shown in FIG. 7, ion implantation is performed on the initial filling layer **61**. Group V elements may be selected and used to perform ion implantation on the initial filling layer **61**. For

example, a phosphorus (P) ion, an arsenic (As) ion, or the like in the group V elements is selected and used. Alternatively, group IV elements are selected and used to perform ion implantation on the initial filling layer **61**. For example, a germanium (Ge) ion or the like in the group IV elements is selected.

(44) In the process of performing ion implantation on the initial filling layer **61**, a dose of ions implanted into the sidewall and the bottom wall of the fourth groove **80** is greater than a dose of ions implanted into the sidewall and the bottom wall of the third groove **70**, that is, doses of ions implanted into different positions and different deposition thicknesses of the initial filling layer **61** are selectively changed, to affect subsequent etching rates of different positions and different deposition thicknesses of the filling layer **60**, for example, in a subsequent etching processing process, an etching rate of the third groove **70** is 10% to 30% faster than an etching rate of the fourth groove **80** by using a same etching processing parameter, to effectively ensure that a size of a formed etching opening **90** is within a preset difference range.

(45) As shown in FIG. **8**, after an ion implantation process performed on the initial filling layer **61** ends, a RTP process is used to process the initial filling layer **61**. RTP can effectively activate an ion implanted in the initial filling layer **61**. An ion doping concentration in the fourth groove **80** is greater than an ion doping concentration in the third groove **70**, such that the etching rate of the fourth groove **80** is slower and the etching rate of the third groove **70** is faster when the initial filling layer **61** is subsequently etched. In the etching process, parameters such as etching time and an etching gas may be appropriately controlled, such that the etching opening **90** whose inner diameter satisfies the target size required by the etching processing process is subsequently formed on the first dielectric layer **20** and/or the filling layer **60**, thereby effectively resolving a problem that LCDU is excessively low in an etching process in the prior art and improving the electrical property of the semiconductor structure.

(46) Referring to FIG. **7**, in some embodiments, to effectively increase a subsequent etching rate in the semiconductor structure, group III elements may be alternatively used to perform ion implantation on the initial filling layer **61**, for example, a boron ion (B) or the like in the group III elements is selected and used. Alternatively, group V elements may be used to perform ion implantation on the initial filling layer **61**. For example, a P ion, an As ion, or the like in the group V elements is selected and used. In an example, when ion implantation is performed on the initial filling layer **61**, an ion implantation dose is between $10 \times 10^{14}/\text{cm}^2$ to $10 \times 10^{15}/\text{cm}^2$, and ion implantation energy is between 10 KeV to 40 KeV.

(47) For etching processing parameters, when an etching rate is 3 nm/min to 4 nm/min, and a deposition thickness of the first dielectric layer **20** is 20 nm to 30 nm, the B ion in the group III elements may be selected and used.

(48) For etching processing parameters, when an etching rate is 4 nm/min to 5 nm/min, and a deposition thickness of the first dielectric layer **20** is 25 nm to 40 nm, the As ion in the group V elements may be selected and used.

(49) For etching processing parameters, when an etching rate is 4 nm/min to 5 nm/min, and a deposition thickness of the first dielectric layer **20** is 30 nm to 50 nm, the P ion in the group V elements may be selected and used.

(50) It should be noted that, after the ion in the group III elements or the group V elements is implanted into the initial filling layer **61**, the dose of ions implanted into the sidewall and the bottom wall of the fourth groove **80** is greater than the dose of ions implanted into the sidewall and the bottom wall of the third groove **70**.

(51) In this embodiment, the ion in the group III elements or the ion in the group V elements is implanted into the initial filling layer. Doses of ions in the group III elements or the group V elements implanted into the different positions of the initial filling layer **61** are different, where an implantation dose in a region (to be specific, the bottom wall and the sidewall of the fourth groove **80**) whose deposition thickness is thicker is greater than an implantation dose in a region (to be

specific, the bottom wall and the sidewall of the third groove **70**) whose deposition thickness is thinner. The doses of ions implanted into the different positions or different deposition thicknesses of the initial filling layer are selectively changed, to help affect subsequent etching rates of the different positions of the initial filling layer.

(52) Referring to FIG. **6**, in some embodiments, the initial filling layer **61** is formed in the first groove **40** and the second groove **50** by using a chemical vapor deposition process or a physical vapor deposition process. A material of the initial filling layer **61** may include but is not limited to silicon oxide.

(53) In the process of forming the initial filling layer **61** by using the deposition processes, the used chemical vapor deposition process or physical deposition process has uneven levels of step coverage, the deposition material forms overhangs at openings of the grooves, and an opening of a groove whose inner diameter is smaller is preferentially closed. Because the inner diameter of the second groove **50** is greater than the inner diameter of the first groove **40**, a deposition thickness of the deposition material for forming the initial filling layer **61** on the sidewall and the bottom wall of the second groove **50** is greater than a deposition thickness on the sidewall and the bottom wall of the first groove **40**. In addition, in the deposition process of the initial filling layer **61**, deposition parameters (such as a deposition rate, deposition time, and particle deposition energy) of the deposition material used for forming the initial filling layer **61** may be controlled, such that a difference between the inner diameters of the third groove **70** and the fourth groove **80** that are formed after the deposition is decreased.

(54) After the deposition is completed, the initial filling layer **61** deposited on a top surface of the mask structure **30** may be polished, to remove the initial filling layer **61** with an uneven thickness at an opening of the first groove **40** and an opening of the second groove **50**, to ensure that when the opening of the first groove **40** is opened, the initial filling layer **61** located at the bottom wall and the sidewall of the first groove **40** and the bottom wall and the sidewall of the second groove **50** is exposed. In this case, the initial filling layer **61** is uniformly distributed on the bottom wall and the sidewall of the first groove **40** and the bottom wall and the sidewall of the second groove **50**.

(55) Referring to FIG. **6**, in some embodiments, the thickness of the initial filling layer **61** is from 5 nm to 8 nm, for example, the thickness of the initial filling layer **81** is 5 nm, 5.5 nm, 6 nm, 6.5 nm, 7 nm, 7.5 nm, 8 nm, or the like.

(56) In this embodiment, the initial filling layer with the thickness of 5 nm to 8 nm is formed, and the implantation doses of the ions in the group III elements or the group V elements are controlled in conjunction with types of the ions, such that the thickness of the initial filling layer deposited on the bottom wall and the sidewall of the second groove **50** is greater than the thickness of the initial filling layer deposited on the bottom wall and the sidewall of the first groove **40**, thereby helping control the etching rates of the third groove **70** and the fourth groove **80** that are subsequently formed.

(57) Referring to FIG. **2**, in some embodiments, a material of the first dielectric layer **20** may include but is not limited to one of silicon oxynitride, silicon nitride, silicon oxide, silicon carbide, and a spin-coated hard mask layer. In some other embodiments, the material of the first dielectric layer **20** may further include undoped polycrystalline silicon. The first dielectric layer **20** is formed by depositing the foregoing deposition materials, and the initial filling layer **61** formed on the first dielectric layer **20** may be formed by depositing silicon oxide. That is, the first dielectric layer **20** and the initial filling layer **61** are formed by depositing different deposition materials. In a chemical vapor deposition process, a thickness of a material deposited at the step of the groove **31** is thicker than a thickness of a material deposited at the bottom of the groove **31**. In this case, in a subsequent process of depositing the initial filling layer **61**, the opening of the first groove **40** with a smaller inner diameter is preferentially closed, such that a deposition material of the initial filling layer **61** in the first groove **40** is fewer, and a deposition material, to be specific, a deposition material of the

initial filling layer **61**, in the second groove **50** with a larger inner diameter is more. Therefore, the thickness of the initial filling layer **61** deposited by using the deposition material for forming the initial filling layer **61** in the second groove **50** with a larger inner diameter is greater than the thickness of the initial filling layer **61** deposited in the first groove **40** with a smaller inner diameter.

(58) After the initial filling layer **61** is formed, the initial filling layer **61** in the first groove **40** forms the third groove **70**, and the initial filling layer **61** in the second groove **50** forms the fourth groove **80**. The deposition thickness of the initial filling layer **61** in the fourth groove **80** is thicker, to enable the second difference between the fourth groove **80** and the third groove **70** to be less than the first difference, such that the etching rates of the different positions of the initial filling layer **61** are subsequently affected by ion implantation and RTP.

(59) It should be noted that, parameters such as a deposition material, ion doping, and a deposition thickness are used in conjunction with each other, and after the third groove **70** and the fourth groove **80** are formed, an appropriate etching selectivity ratio is selected between the mask structure **30** and the filling layer **60**, where the etching selectivity ratio ranges from 1:3 to 1:6, such that in a subsequent etching processing process, an etching rate of the filling layer **60** in the third groove **70** is faster and an etching rate of the filling layer **60** in the fourth groove **80** is slower. The following cases are included.

(60) After the filling layer **60** in the third groove **70** is completely removed through etching, etching is stopped. In this case, the filling layer **60** in the fourth groove **80** is not completely removed, and the inner diameter of the fourth groove **80** is decreased due to the existence of the filling layer **60**. In this case, if the inner diameter of the fourth groove **80** and the inner diameter of the third groove **70** from which the filling layer **60** is removed satisfy a preset difference, subsequent etching is performed by using the fourth groove **80** and the sidewall of the third groove **70** as a mask, to further ensure subsequent formation precision of a first through hole **130**, a second through hole **140**, and a contact hole **150** (referring to FIG. **11** to FIG. **13**)

(61) After the filling layer **60** of the fourth groove **80** is completely removed through etching, the filling layer **60** in the third groove **70** is already completely removed. In this case, the first dielectric layer **20** on a side of the third groove **70** is further etched and part of the first dielectric layer **20** is removed, to ensure that a plurality of etching openings **90** whose inner diameters have a preset difference are subsequently formed, thereby ensuring subsequent formation precision of the first through hole **130**, the second through hole **140**, and the contact hole **150** (referring to FIG. **11** to FIG. **13**).

(62) The etching openings **90** whose inner diameters have a preset difference specifically determine etching time of the third groove **70** and the fourth groove **80** according to a target size.

(63) Referring to FIG. **2**, in some embodiments, the base **10** may include a substrate **11** and a support structure **12** provided on the substrate **11**. In this embodiment, the base **10** may be formed by using the following method.

(64) The substrate **11** is provided, where the substrate **11** may be made of a semiconductor material. The semiconductor material may be one or more of silicon, germanium, silicon-germanium, and a silicon-carbon compound.

(65) The support structure **12** is formed on a top surface of the substrate **11** by using an atomic layer deposition process, a physical vapor deposition process, or a chemical vapor deposition process. It should be noted that the support structure **12** may be a single-layer structure or a stacked structure. When the support structure **12** is a stacked structure, the support structure **12** may include a first support layer, a second support layer, and the like that are stacked. The support structure **12** may be configured to form through holes to be formed in a subsequent manufacturing process, where the through hole may include a contact hole configured to form a contact structure, a capacitor hole configured to form a capacitor structure, a bit line contact hole configured to form a bit line, and the like. After the through holes are formed, the support structure **12** provided around

the through holes may be further configured to subsequently form another semiconductor structure between adjacent through holes, such as a bit line structure, a metal wire layer, or the like.

(66) Referring to FIG. 2, in some embodiments, a plurality of isolation structures **111** provided at intervals may be provided in the substrate **11** along a first direction X.

(67) A process of forming the isolation structure **111** may use the following method.

(68) A photoresist layer (not shown in the figure) and a mask layer (not shown in the figure) that are stacked are deposited on a top surface of the substrate **11** by using an atomic layer deposition process, a physical vapor deposition process, or a chemical vapor deposition process. A mask pattern is formed on the photoresist layer through exposure or development etching. The photoresist layer with the mask pattern is used as a mask to etch and remove part of the photoresist layer and part of the mask layer. In this way, a plurality of isolation trenches (not shown in the figure) provided at intervals along the first direction X are formed on the substrate **11**.

Subsequently, the remaining photoresist layer and mask layer are removed through etching. Then, the isolation structure **111** is formed through deposition in the isolation trench by using the atomic layer deposition process, the physical vapor deposition process, or the chemical vapor deposition process. The substrate **11** between adjacent isolation structures **111** forms an active region **112**. The isolation structure **111** may be made of an insulating material, such as silicon dioxide or silicon oxynitride. The plurality of isolation structures provided at intervals are used to form active regions provided at intervals on the substrate, to help subsequently form other structures connected to the active regions on the substrate, such as a capacitor contact, a source line contact, and a bit line contact.

(69) Referring to FIG. 2, the support structure **12** is a stacked structure, where the support structure **12** may be formed by using the following method.

(70) An isolation layer **121**, a first sacrificial layer **122**, at least one support layer **123**, a second sacrificial layer **124**, and an etching layer **125** that are sequentially stacked are formed on the substrate **11** by using the atomic layer deposition process, the physical vapor deposition process, or the chemical vapor deposition process. A material of the isolation layer **121** may include but is not limited to silicon nitride, materials of the first sacrificial layer **122** and the second sacrificial layer **124** may include but are not limited to silicon dioxide, and the etching layer **125** may include but is not limited to a spin-coated hard mask layer.

(71) The support layer **123** includes a first support layer **1231** and a spin-coated hard mask layer **1232** that are alternately arranged. A material of the first support layer **1231** may include but is not limited to silicon oxynitride. In this example, there are two support layers **123**, to be specific, the two support layers **123** are provided between the first sacrificial layer **122** and the second sacrificial layer **124**. It should be noted that, in the two support layers **123**, a thickness of an upper spin-coated hard mask layer **1232** may be equal to a thickness of a lower spin-coated hard mask layer **1232**, or a thickness of an upper spin-coated hard mask layer **1232** is greater than or less than a thickness of a lower spin-coated hard mask layer **1232**.

(72) In this embodiment, stacked support structures are used, to precisely transfer etching sizes of etching openings onto the support structures in sequence, such that molding precision for subsequently forming other semiconductor structures (a bit line contact structure, a capacitor contact structure, and the like) in the support structures is improved, thereby improving a yield of the semiconductor structures.

(73) Referring to FIG. 2 to FIG. 5, in some embodiments, the mask structure **30** may be formed by using the following method.

(74) As shown in FIG. 2, after the etching layer **125** is formed, the first dielectric layer **20** is formed on the etching layer **125** by using the atomic layer deposition process, the physical vapor deposition process, or the chemical vapor deposition process.

(75) Referring to FIG. 3 and FIG. 4, a first mask layer **100** is formed on the first dielectric layer **20** by still using the deposition processes.

(76) Subsequently, a first mask pattern **101** is provided, and part of the first mask layer **100** is removed by using an etching processing process based on the first mask pattern **101**, to form a plurality of initial grooves **102** on the first mask layer **100**.

(77) Part of the first dielectric layer **20** is removed by using the remaining part of the first mask layer **100** as a mask based on the initial grooves **102**, to form the grooves **31** on the first dielectric layer **20**. The initial grooves **102** may include a plurality of fifth grooves **110** and a plurality of sixth grooves **120**, and an inner diameter of the fifth groove **110** is different from an inner diameter of the sixth groove **120** along the first direction X.

(78) Then, part of the first dielectric layer **20** is removed based on the fifth groove **110** by using the etching processing process, to form the first groove **40** on the first dielectric layer **20**. In addition, part of the first dielectric layer **20** is removed based on the sixth groove **120** by using the etching processing process, to form the second groove **50** on the first dielectric layer **20**. The remaining part of the first dielectric layer **20** forms a plurality of mask structures **30** (referring to FIG. 5) provided at intervals along the first direction X.

(79) In this embodiment, the first groove and the second groove can be quickly formed on the first dielectric layer by using the first mask pattern. A method is simple and is easily to operate.

(80) Referring to FIG. 11 to FIG. 13, in some embodiments, the method of manufacturing a semiconductor structure includes:

(81) As shown in FIG. 11, part of the etching layer **125** is removed by using the etching processing process based on etching openings **90** satisfying a target size requirement, to form a plurality of first through holes **130** arranged in an array in the etching layer **125**. After the first through holes **130** are formed, the remaining part of the first dielectric layer **20** on the top of the etching layer **125** continues to be removed by using a polishing processing process.

(82) As shown in FIG. 12, after the first dielectric layer **20** is removed, part of the second sacrificial layer **124** is removed by using the etching processing process based on the first through holes **130**, to form a plurality of second through holes **140** arranged in an array in the second sacrificial layer **124**. After the second through holes **140** are formed, the remaining part of the etching layer **125** continues to be removed by using the etching processing process.

(83) As shown in FIG. 13, after the etching layer **125** is removed, part of the support structure **12** is removed by using the etching processing process based on the second through holes **140**, to form a plurality of contact holes **150** arranged in an array in the support structure **12**. The contact holes **150** expose at least part of the active region **112**.

(84) In this embodiment, the etching openings and the first through holes are used, to effectively improve formation precision of the second through holes, and further help subsequently form other semiconductor structures, such as a bit line contact structure and a capacitor contact structure, in the support structure.

(85) As shown in FIG. 1, an exemplary embodiment of the present disclosure provides a semiconductor structure. The semiconductor structure may be manufactured by using the method of manufacturing a semiconductor structure according to any one of the foregoing embodiments. The semiconductor structure may be a DRAM, and certainly, the semiconductor structure may be further another structure.

(86) It should be noted that in the description of the present disclosure, the terms such as “center”, “top”, “bottom”, “left”, “right”, “vertical”, “horizontal”, “inner” and “outer” indicate the orientation or position relationships based on the accompanying drawings. These terms are merely intended to facilitate description of the present disclosure and simplify the description, rather than to indicate or imply that the mentioned apparatus or element must have a specific orientation and must be constructed and operated in a specific orientation. Therefore, these terms should not be construed as a limitation to the present disclosure.

(87) It can be understood that the terms such as “first” and “second” used in the present disclosure can be used to describe various structures, but these structures are not limited by these terms.

Instead, these terms are merely intended to distinguish one structure from another.

(88) The same elements in one or more accompanying drawings are denoted by similar reference numerals. For the sake of clarity, various parts in the accompanying drawings are not drawn to scale. In addition, some well-known parts may not be shown. For the sake of brevity, a structure obtained by implementing a plurality of steps may be shown in one figure. In order to understand the present disclosure more clearly, many specific details of the present disclosure, such as the structure, material, size, processing process, and technology of the device, are described below. However, as those skilled in the art can understand, the present disclosure may not be implemented according to these specific details.

(89) Finally, it should be noted that the above embodiments are merely intended to explain the technical solutions of the present disclosure, rather than to limit the present disclosure. Although the present disclosure is described in detail with reference to the above embodiments, those skilled in the art should understand that they may still modify the technical solutions described in the above embodiments, or make equivalent substitutions of some or all of the technical features recorded therein, without deviating the essence of the corresponding technical solutions from the scope of the technical solutions of the embodiments of the present disclosure.

Claims

1. A method of manufacturing a semiconductor structure, comprising: providing a base; forming a first dielectric layer on the base, wherein the first dielectric layer is provided with grooves arranged in an array, the grooves comprise a first groove and a second groove, and an inner diameter of the first groove and an inner diameter of the second groove have a first difference; forming an initial filling layer, wherein the initial filling layer covers a sidewall and a bottom wall of the first groove and covers a sidewall and a bottom wall of the second groove, the initial filling layer in the first groove forms a third groove, the initial filling layer in the second groove forms a fourth groove, an inner diameter of the third groove and an inner diameter of the fourth groove have a second difference, and the second difference is less than the first difference; processing the initial filling layer to form a filling layer; and etching at least one of the filling layer or the first dielectric layer, to form a plurality of etching openings whose inner diameters have a preset difference on a surface of the base; wherein the processing the initial filling layer to form a filling layer comprises: performing ion implantation on the initial filling layer; and performing rapid thermal processing on the initial filling layer after the ion implantation, to form the filling layer.

2. The method of manufacturing a semiconductor structure according to claim 1, wherein the groove has a target size, the inner diameter of the first groove is less than the target size, and the inner diameter of the second groove is greater than the target size; the target size is any value in a range of 30 nm to 60 nm; and the inner diameter of the first groove and the inner diameter of the second groove have a maximum difference of 6 nm.

3. The method of manufacturing a semiconductor structure according to claim 1, wherein the performing ion implantation on the initial filling layer comprises: performing ion implantation on the initial filling layer by using a group III or group V element.

4. The method of manufacturing a semiconductor structure according to claim 3, wherein an ion implantation dose is between $10 \times 10^{14} / \text{cm}^2$ and $10 \times 10^{15} / \text{cm}^2$, and ion implantation energy is between 10 KeV and 40 KeV.

5. The method of manufacturing a semiconductor structure according to claim 1, wherein the forming an initial filling layer comprises: forming the initial filling layer in the first groove and the second groove by using a chemical vapor deposition process or a physical vapor deposition process.

6. The method of manufacturing a semiconductor structure according to claim 5, wherein a thickness of the initial filling layer is from 5 nm to 8 nm.

7. The method of manufacturing a semiconductor structure according to claim 1, wherein a material of the first dielectric layer comprises one of silicon oxynitride, silicon nitride, silicon oxide, silicon carbide, and a spin-coated hard mask layer.
8. The method of manufacturing a semiconductor structure according to claim 1, wherein the first dielectric layer comprises undoped polysilicon.
9. The method of manufacturing a semiconductor structure according to claim 1, wherein the providing a base comprises: providing a substrate; and forming a support structure on the substrate, wherein the support structure comprises: an isolation layer, a first sacrificial layer, at least one support layer, a second sacrificial layer, and an etching layer that are formed and sequentially stacked on the substrate, and each support layer comprises a first support layer and a spin-coated hard mask layer that are alternately arranged.
10. The method of manufacturing a semiconductor structure according to claim 9, wherein the forming a first dielectric layer on the base, wherein the first dielectric layer is provided with grooves arranged in an array comprises: forming the first dielectric layer on the etching layer; forming a first mask layer on the first dielectric layer; providing a first mask pattern, and removing part of the first mask layer based on the first mask pattern, to form a plurality of initial grooves on the first mask layer; and removing part of the first dielectric layer by using a remaining part of the first mask layer as a mask based on the initial grooves, to form the grooves on the first dielectric layer.
11. The method of manufacturing a semiconductor structure according to claim 10, further comprising: removing part of the etching layer based on the etching openings, to form a plurality of first through holes arranged at intervals along a first direction in the etching layer.
12. The method of manufacturing a semiconductor structure according to claim 11, further comprising: removing part of the second sacrificial layer based on the first through holes, to form a plurality of second through holes arranged at intervals along the first direction in the second sacrificial layer.
13. The method of manufacturing a semiconductor structure according to claim 12, further comprising: forming, based on the second through holes, a plurality of contact holes arranged at intervals along the first direction in the support structure.
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